

DistrictDx

District-Level Pharmaceutical Market Attractiveness Index

Supporting Technical Note

Trilytics 2026 - Sun Pharmaceutical Industries × IIM Calcutta PGDBA Conclave

Executive Summary

India's pharmaceutical territory planning relies on backward-looking inputs - state aggregates, historical sales, and regional manager intuition - masking tenfold variation across districts and rewarding past allocation over future potential.

DistrictDx replaces them with an evidence-based framework scoring all 785 districts - built entirely from free public data, fully auditable, and reproducible by any analyst.

Dimension	Definition	Commercial Meaning
Demand Potential	Epidemiological, demographic, and economic capacity to generate therapy need	"How large is the market opportunity?"
Realizability	Healthcare infrastructure, access, and affordability to convert need into prescriptions	"Can we actually reach this market?"

The index is the **geometric mean** of both axes - not a weighted sum. High demand with negligible infrastructure scores low because that demand is currently uncapturable.

The framework produces **three indices per district**: MAI_Overall (portfolio allocation), MAI_Chronic (diabetes, hypertension, cardiovascular), MAI_Acute (infections, child health), plus a Future Opportunity Index.

Constructed from **19 variables across 11+ public datasets**, the framework applies hierarchical imputation, redundancy-aware factor analysis, AHP weighting ($CR < 0.1$), and within-state quadrant

classification. Validation uses two independent mechanisms: Spearman correlation against external proxies (NSSO OOP $\rho = 0.73$, Jan Aushadhi $\rho = 0.45$, PMJAY $\rho = 0.51$) and entropy-weighting sensitivity (AHP vs. data-driven, $\rho \geq 0.84$).

Key Takeaway - DistrictDx converts fragmented public data into actionable district-level intelligence, replacing intuition with evidence at national scale.

Why DistrictDx?

Dimension	Traditional Approach	DistrictDx
Geographic resolution	State aggregates	District scores (785 districts)
Temporal orientation	Backward-looking (past sales)	Forward-looking (current + trend)
Prioritization basis	Manager intuition	Explainable statistical framework
Therapy granularity	Uniform planning	Therapy-specific (Overall, Chronic, Acute)
Weighting rationale	Implicit or arbitrary	AHP-derived, literature-grounded, verified
Data limitations	Hidden in aggregates	Explicit confidence scores per district
Reproducibility	Tribal knowledge	Full pipeline reproducible from source
Decision framework	Single rank-order list	2×2 quadrant → distinct commercial playbooks

Key Takeaway - DistrictDx moves territory planning from intuition and aggregation to evidence and granularity.

Framework Overview

- 1. Data Foundation.** Eleven public datasets assembled - Census, NFHS-5/4, NASA VIIRS, Rural Health Statistics, PMGSY, NVBDCP/IDSP, LGD codes. All free; no proprietary input.
- 2. Administrative Reconciliation.** Every district receives a canonical LGD crosswalk key. Post-2011 boundary changes reconciled via concordance analysis; inherited districts flagged, none dropped.
- 3. Data Quality & Estimation.** Variables winsorized at 1st and 99th percentiles. Missing values - NFHS-5 covers 707 of 785 districts - filled via four tiers: district observation → NFHS-4 trend-adjusted → state average → national average. Each imputation flagged. Every district receives a **confidence score** (fraction of variables directly observed).
- 4. Variable Weighting.** Variables grouped into four domains (Demand-Chronic, Demand-Acute, Realizability-Chronic, Realizability-Acute), min-max normalized to [0, 1], checked for pairwise redundancy ($|r| > 0.8 \rightarrow$ Factor Analysis). AHP weights derived from literature-grounded score vectors, verified for consistency ($CR < 0.1$).
- 5. Index Construction.** Domain composites blended into Demand and Realizability axes (50/50 chronic-acute default). Final MAI is the geometric mean:
$$MAI = Demand^{\alpha} \times Realizability^{(1-\alpha)} \quad (\text{default } \alpha = 0.5)$$

Three variants: Overall, Chronic, Acute. Blend ratio adjustable by portfolio mix.
- 6. Validation.** AHP weights compared against Shannon entropy weights (Spearman $\rho \geq 0.84$). Index correlated against NSSO OOP, Jan Aushadhi, HMIS, and PMJAY.
- 7. Future Projection.** Historical baseline reconstructed from NFHS-4 (~2015) using identical normalization and weights. Trend slopes dampened ($\beta = 0.3$, sensitivity-tested at $\beta \in \{0.2, 0.3, 0.4\}$) to produce Future MAI.

Key Takeaway - Every stage serves a purpose: reconciliation prevents boundary data loss, imputation respects missingness, redundancy checks prevent double-counting.

Scientific Rationale

Why Demand × Realizability

Market attractiveness is two-dimensional. Demand without infrastructure is uncapturable; infrastructure without demand is oversupply. Either axis near-zero yields a near-zero score.

Commercial implication: High diabetes prevalence with no endocrinologists, few pharmacies, and low income is not an attractive market today - regardless of patient population.

Scenario	Demand	Realizability	Linear Sum ($\alpha=0.5$)	Geometric Mean ($\alpha=0.5$)	Correct Signal
Balanced market	0.5	0.5	0.50	0.50	✓
High demand, low access	0.9	0.1	0.50	0.30	✓ - penalizes imbalance
Low demand, high access	0.1	0.9	0.50	0.30	✓ - penalizes imbalance
Moderate both	0.7	0.5	0.60	0.59	✓

This two-axis structure maps directly to resource allocation. Star markets (high demand, high realizability) are ready for sales-force deployment. Emerging markets (high demand, low realizability) need market-development investment first - two distinct playbooks, not a single rank order.

Key Takeaway - Two axes prevent a single high score from masking a critical weakness - exactly what a linear composite would do.

Why Geometric Mean Instead of Weighted Sum

A weighted sum lets one high axis compensate for a near-zero other. At Demand = 0.9 and Realizability = 0.1, a weighted sum ($\alpha = 0.5$) yields 0.50 - "moderate attractiveness" where 90% of demand cannot be reached. The geometric mean yields **0.30**.

Property	Weighted Sum	Geometric Mean	Business Rationale
D=0.9, R=0.1	0.50 (misleading)	0.30 (correct)	Low access drags score down

Property	Weighted Sum	Geometric Mean	Business Rationale
Formula	$\alpha \cdot D + (1-\alpha) \cdot R$	$D^\alpha \times R^{(1-\alpha)}$	Non-linear penalty for imbalance
Sensitivity	Linear	Sub-linear	Improving the weak axis matters more

Alpha sensitivity tested at 0.4, 0.5, 0.6. Rank correlation exceeds 0.98 across all values - ranking is robust.

Key Takeaway - The geometric mean requires both axes to score well, matching the commercial reality that demand without access is uncapturable.

Why AHP Weighting

Approach	Why Not Used
Equal weighting	Implies sex ratio matters as much as diabetes prevalence - overweights marginal variables
Data-driven (PCA, entropy)	Maximizes variance, not business importance. Low-variance variables get zero weight regardless of commercial relevance
AHP (selected)	Literature-grounded pairwise comparisons, consistency-verified (CR < 0.1), auditable, tested against entropy weights ($\rho \geq 0.84$)

AHP produces weights any sales manager can understand and challenge - each traces to documented comparisons, with a consistency check (CR < 0.1) that halts construction if too contradictory. The $\rho \geq 0.84$ correlation with entropy weights confirms rankings are not artifacts of subjective judgment.

Key Takeaway - AHP balances expert judgment with mathematical rigor; entropy sensitivity ($\rho \geq 0.84$) confirms weights are defensible, not arbitrary.

Additional Decisions

Decision	Problem It Addresses	Solution	Business Impact
Confidence scores	78 of 785 districts lack direct NFHS-5 data	1 – imputed fraction; low-confidence districts flagged visually	Prevents imputed districts appearing observed; enables risk-calibrated investment
Hierarchical imputation	Missingness not random - remote districts underrepresented	Four tiers: district → NFHS-4 trend-adjusted → state → national	Preserves district variation; degrades gracefully
Within-state benchmarking	National medians favor wealthy states, producing a GDP map	Each district compared to within-state median only	Reflects actual pharma deployment patterns
Proxy validation	District-level sales proprietary; calibration impossible	Rank correlation: NSSO OOP ($\rho=0.73$), Jan Aushadhi ($\rho=0.45$), PMJAY ($\rho=0.51$)	Face validity without overclaiming; non-significant results disclosed
Explainable framework	ML models uninterpretable; risk of overfitting to proxies	Transparent AHP with documented weights, flagged imputations, reproducible pipeline	A sales manager can understand any district's score

Key Takeaway - Every methodological choice serves a purpose: transparency, defensibility, and practical utility for pharmaceutical decisions.

Validation Evidence

Test	Spearman ρ	p-corrected	Interpretation
AHP vs Entropy (Overall)	0.918	-	Strong rank agreement - index robust to weighting method
AHP vs Entropy (Chronic)	0.912	-	Strong rank agreement
AHP vs Entropy (Acute)	0.842	-	Good rank agreement
MAI $\times$ NSSO OOP/capita	0.728	0.0000	Strong - realized healthcare purchasing
MAI_Chronic $\times$ NSSO OOP/capita	0.782	0.0000	Strong - chronic drives out-of-pocket spending
MAI $\times$ Jan Aushadhi density	0.455	0.0000	Moderate - public pharmacy penetration
MAI_state $\times$ PMJAY vol/1k	0.509	0.0015	Moderate - state insurance utilization
MAI $\times$ HMIS OPD/capita	0.036	1.0000	Not significant (expected - HMIS quality)

Both mechanisms converge: the index captures real variation in healthcare market activity. The strongest signal - NSSO OOP ($\rho = 0.73$) - confirms higher-MAI districts exhibit higher realized healthcare purchasing, the closest available proxy for pharmaceutical consumption.

Key Takeaway - Entropy sensitivity and proxy correlation converge: the index captures real market variation, not artifacts of expert judgment.

Business Interpretation

From Scores to Decisions

A numerical MAI score is necessary for ranking but insufficient for resource allocation. The quadrant classification maps directly to investment strategy:

Quadrant	Demand	Realizability	Signal	Commercial Action
Star Market	High	High	Both dimensions strong	Deploy, defend, prioritize new launches
Emerging	High	Low	Demand exists, access limits conversion	Invest in pharmacy networks and hospital partnerships before scaling sales
Underserved	Low	High	Infrastructure ahead of need	Maintain coverage, monitor for demand shifts
Deprioritize	Low	Low	Both dimensions weak	Monitor; revisit when conditions shift

A Star in Bihar may score lower absolutely than an Emerging district in Maharashtra - but relative to its peers, it is the priority for a regional manager with a fixed budget.

Therapy-Specific Strategy

Index	Use Case	Example Portfolio
MAI Overall	Portfolio-level territory allocation	Cross-therapy resource planning
MAI Chronic	Chronic franchise strategy	Diabetes, hypertension, cardiovascular
MAI Acute	Acute franchise strategy	Anti-infectives, pediatric, vaccines

A district aging and urban may rank high on Chronic but low on Acute - franchise-specific scores prevent misallocation.

Future Opportunity

The Future Index answers: *"Where is this district heading?"*

Gain Signal	Interpretation	Commercial Implication
High positive gain	District improving rapidly	Early investment ahead of competitors
High current MAI, low gain	Mature market, plateaued growth	Defend share, optimize cost
Negative gain	Declining attractiveness	Monitor, consider reallocation

$\beta = 0.3$ dampens the four-year trend to 30% extrapolation. Government interventions (Swachh Bharat, Jal Jeevan) may accelerate trends; NFHS-5 methodological changes may not sustain.

Key Takeaway - The framework transforms statistical outputs into three concrete commercial decisions: where to deploy, where to develop, and where to wait.

Limitations & Reproducibility

Limitation	Impact	Mitigation
Census 2011 data is 15+ years old	Demographics may have shifted	Most comprehensive baseline until Census 2021
NFHS-5 covers 707 of 785 districts	78 rely on imputed values	Confidence scores flag every estimated district
Nightlights capture formal economy only	Rural subsistence undercounted	Disclosed; no superior public alternative exists
No ground-truth pharmaceutical sales	Cannot calibrate against actual prescriptions	Proxy validation (NSSO OOP $\rho = 0.73$)
Within-state splits obscure cross-state variation	Bihar Star < Maharashtra Emerging absolutely	Supplement with absolute scores

Reproducibility & Extensibility

Every output independently reproducible from source - no proprietary data, no API keys, no black-box steps. Every imputation flagged, every weight documented. The framework extends to new therapy categories without code changes; new data sources integrate via the data dictionary.

Conclusion

DistrictDx provides an **explainable, reproducible, and commercially meaningful** framework for district-level pharmaceutical planning.

What DistrictDx Is	What DistrictDx Is Not
Evidence-based, auditable, and transparent	A black-box algorithm
Validated against available proxies with disclosed limitations	A "validated" model with ground-truth calibration
Designed for strategic resource allocation decisions	A predictive engine for sales forecasting
Reproducible by any analyst from public data	Dependent on proprietary or purchased datasets

DistrictDx answers a question historically unanswerable at scale: *"Across 785 districts, which ones deserve my sales force today - and which should I develop for tomorrow?"*

The methodology is deliberately transparent - every weight challengeable, every imputation flagged, every conclusion traceable. It is not the final word, but a significant improvement over the state aggregates, historical sales, and intuition that currently guide allocation.

Key Takeaway - DistrictDx makes district-level pharmaceutical intelligence auditable, reproducible, and actionable - 19 public variables into a defensible framework for 785 districts.

Project Resources

Interactive Dashboard: <https://districtdx.vercel.app/>

GitHub Repository: <https://github.com/karbburn/DistrictDx>

The GitHub repository contains the complete implementation, preprocessing pipeline, validation methodology, and documentation. The interactive dashboard enables exploration of district-level pharmaceutical market attractiveness across all 785 Indian districts.

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